

Central Banks in Times of Crypto, Crisis, and Change

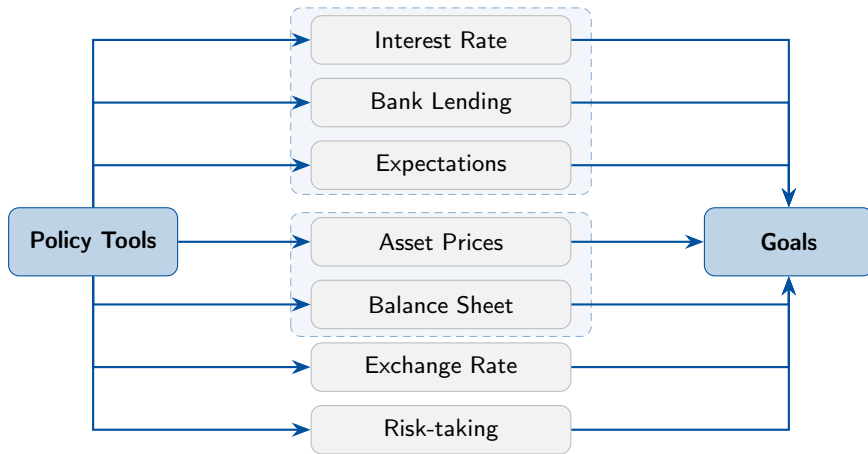
Alexander Dentler

If money moves from bank deposits to crypto —
what exactly breaks in central banking?

By the end of the next 30 minutes, you can:

- 1 **Differentiate** cryptocurrency, stablecoins, and bank deposits.
- 2 **Explain** how stablecoins can weaken at least two monetary-transmission channels.
- 3 **Compare** MiCAR (EU) and the GENIUS Act (US) as two regulatory responses — and understand why the choice is also industrial policy.

The Transmission Mechanism — Seven Channels



Known:

What changed:

To solve:

CB steers the economy through these seven channels.
private digital money and competing monetary jurisdictions.
which channels actually weaken — and which only appear to?

What Is Crypto?

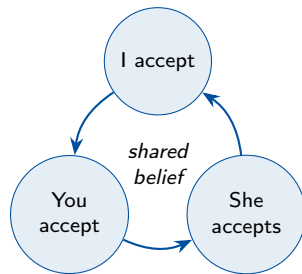
CRYPTO

A digital asset recorded on a distributed ledger, secured by cryptography, transferable peer-to-peer without a trusted intermediary.

- Money looks like a close analogue

MONETARY VALUE

A valuation derived from the **shared belief** that others will accept it in the future.

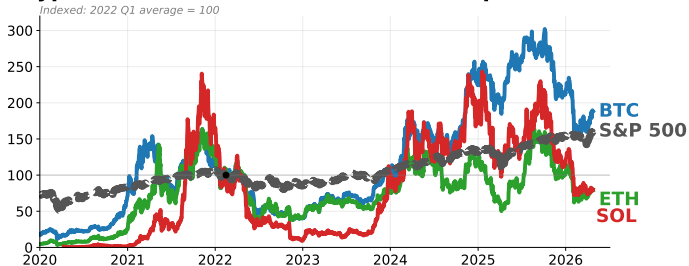


Crypto Currencies vs. S&P 500 — No Fundamentals, No Gravity

NATIVE (NON-CLAIM) CRYPTOCURRENCY

A cryptocurrency with protocol-based issuance and **no redemption claim** on any issuer.

Crypto Currencies vs. S&P 500, 2020-Apr 2026



Source: Yahoo Finance via yfinance; daily adjusted closes. S&P 500 non-trading days forward-filled.

Underlying stocks are anchored in expected earnings, shaped by central bank policy.
Crypto escaped the earnings anchor by design. The volatility is the cost.

Why Markets in Crypto Are Different From Other Markets

- **Observation:** multiple competing currencies with volatile valuations.
- Where have we seen this before?
⇒ FX markets.

Kareken & Wallace (1981, QJE)

With multiple fiat monies in one goods market and **no fundamental value anchor**, the system admits **multiple equilibria**.

Classroom derivation

Single currency

$$\underbrace{M}_{\text{CB fixes}} \cdot \underbrace{V}_{\approx \text{const}} = \underbrace{P}_{\text{real output}} \cdot \underbrace{Y}_{\text{real output}} \implies P = \frac{MV}{Y}$$

Unknown: price level. **Exogenous:** non-monetary.

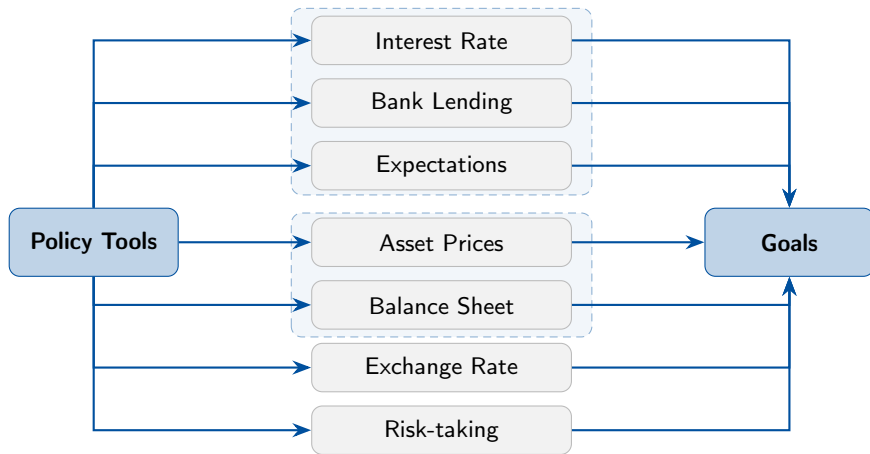
One equation, one unknown.

Two currencies, same goods market

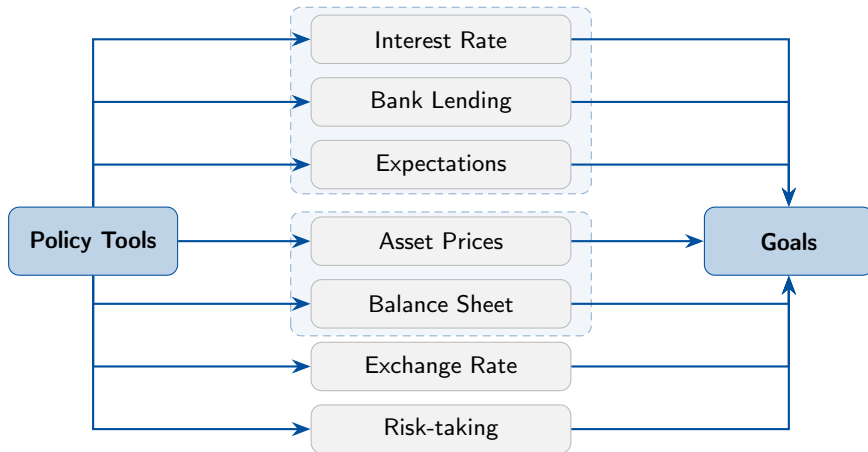
$$(M_{\$} + e \cdot M) \cdot V = P \cdot Y$$

One equation, **two unknowns** (P, e):
for every exchange rate $e \geq 0$ there exists a P that satisfies this.

So: what are we losing?



Cryptocurrency is essentially a store of value and a payment vehicle



Global crypto $\sim \$3\text{ T}$ vs. US household wealth $\sim \$176\text{ T} \Rightarrow <2\%$; true US exposure $\approx 0.5\%$. *Too small in steady state for a channel-level claim.*

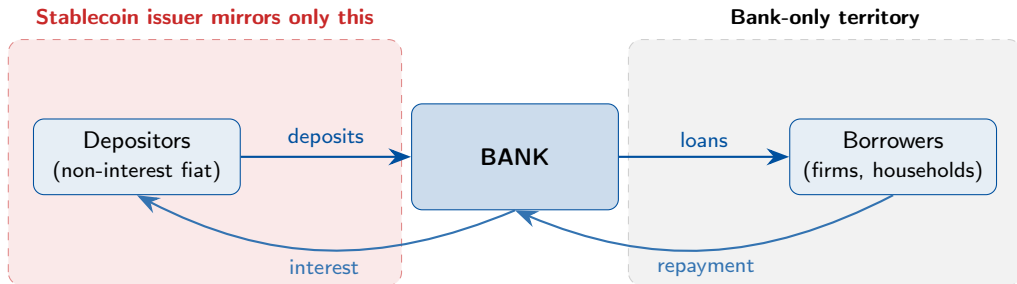
Stablecoins look like Deposit Contracts

STABLECOIN

A *cryptocurrency* whose **issuer** promises **par redemption**, backed by reserve assets.

DEPOSIT CONTRACT

An agreement between a bank and a depositor under which the depositor surrenders funds to the bank in exchange for the right to **withdraw a specified amount on demand at par**.

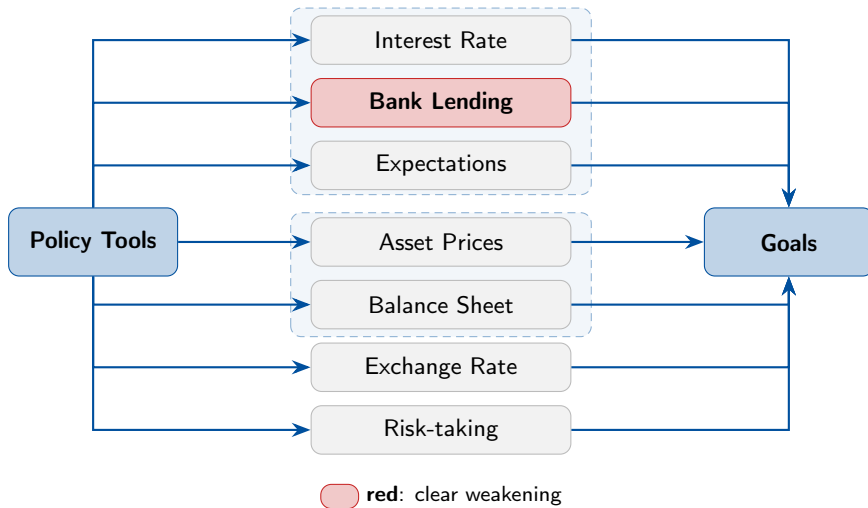


If households hold stablecoins, where does bank funding go?

Household	Commercial bank	Stablecoin issuer																						
<table><tr><th colspan="2"></th></tr><tr><th>Assets</th><th>Liab. (net worth)</th></tr><tr><td>Deposit ↓↓</td><td></td></tr><tr><td>Stablecoin ↑↑</td><td></td></tr></table>			Assets	Liab. (net worth)	Deposit ↓↓		Stablecoin ↑↑		<table><tr><th colspan="2"></th></tr><tr><th>Assets</th><th>Liab.</th></tr><tr><td>Loans ↓↓</td><td>Deposit ↓↓</td></tr><tr><td>Reserves</td><td></td></tr></table>			Assets	Liab.	Loans ↓↓	Deposit ↓↓	Reserves		<table><tr><th colspan="2"></th></tr><tr><th>Assets</th><th>Liab.</th></tr><tr><td>T-bills ↑↑</td><td>Stablecoin ↑↑</td></tr></table>			Assets	Liab.	T-bills ↑↑	Stablecoin ↑↑
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Tether and USDC together held >170bn of Treasury bills by late 2025

Deposit substitution → the Bank-Lending channel



Bank Runs: A Self-Reinforcing Prophecy

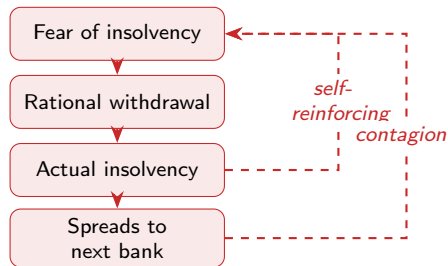
Scene from Mary Poppins (1964):

- 1 Banker brings son to work
- 2 Bank director talks son into investment (via song)
- 3 Son wants his coin back
- 4 Bank director refuses
- 5 Customers overhear, panic, and withdraw their money

BANK RUN

A **bank run** is a rapid, large-scale withdrawal of deposits driven by depositors' fear that the bank cannot honour its claims at par.

- **Diamond & Dybvig (1983):** two equilibria — trust (no run) or panic (run). Both rational.
- Contagion: one bank's run triggers fear at the next.



Modern examples: SVB (2023),
Credit Suisse (2023)

— digital runs in *hours* in countries with
established banking regulations.

Native Crypto vs Stablecoin vs Bank Deposit

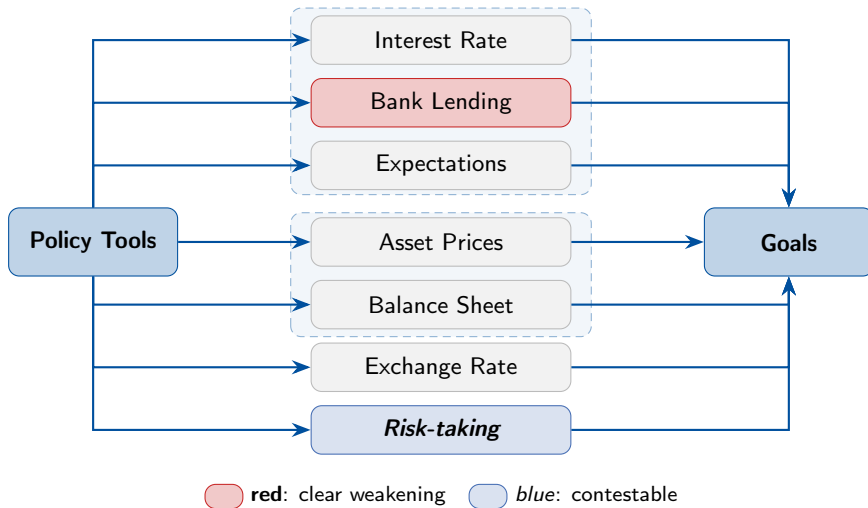
	Native (non-claim) Cryptocurrency	Stablecoin	Bank Deposit
Issuer	Decentralized	Private firm	Commercial bank
Redemption	None	1:1 on demand	1:1 on demand
Backing	None	Reserves (cash, bonds)	Reserves+Loans
Stability	Volatile	Pegged (if reserves hold)	Par (if bank solvent)
Safeguards	None	None	• Prudential liquidity requirements (reserve + liquidity buffers) • Liquidity ratios (LCR/NSFR) • Capital adequacy (Basel III) • Deposit insurance • Lender of last resort
Examples	Bitcoin, Ether	Tether, USDC, EURC	Checking accounts

Tested in stress: **Terra USD, May 2022** — algorithmic, no real reserves — ~\$40–50 bn destroyed in days.

History Rhymes: Free Banking (1830s–1860s) and Stablecoins

	US Free Banking (1837–1863)	Stablecoins Today
Issuance	Any chartered bank issues its own notes (state-chartered, e.g. Tennessee, Illinois, Michigan).	Any firm can issue tokens — e.g. Tether , Circle , Paxos, First Digital.
BIS triple test — both eras fail on all three:		
Singleness ✗	Notes at varying discounts by issuer.	1 USDT $\neq$ 1 USDC in a crisis — depegging.
Elasticity ✗	No LOLR; reserve base couldn't expand with demand.	No LOLR; reserves fixed — can't flex under stress.
Integrity ✗	Frequent failures, forgery, opaque reserves; holders bore losses.	Opaque reserves, uneven audits, cross-border anti-money laundering gaps.
Resolution	National Banking Acts (1863–64): federal charter, uniform currency.	MiCAR / GENIUS Act = the modern equivalent?

Risky storage and the Risk-Taking channel



The Cockroach Problem: Opacity as Crisis Amplifier

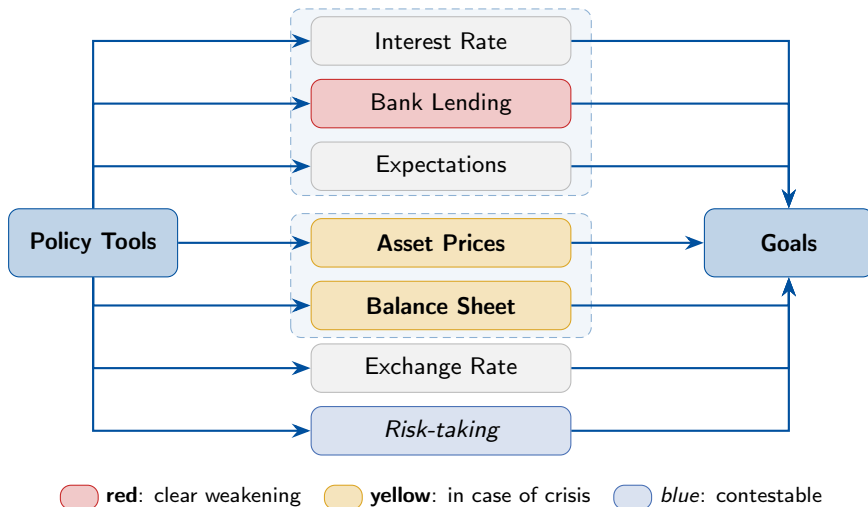
2008: Crisis was *triggered* by bad mortgage loans — but **opacity** turned a bad-loan problem into a global market freeze.

- ❶ Subprime mortgages resold as ABS
- ❷ Nobody knew who held the risk
- ❸ When Lehman fell, interbank lending froze:
“If one cockroach exists, there are more.”
Jamie Dimon, CEO of JPMorgan Chase
- ❹ Rating agencies missed it. Regulators missed it.
⇒ The plumbing was opaque.

2008	Now
Bad subprime loans	Stablecoin reserves + private credit
Packaged in ABS	Issued as tokens
Rating agencies missed opacity	Reserve audits often delayed
Contagion via interbank freeze	Contagion via stablecoin run / T-bill fire-sale

SZ, 24.3.2026.

Fire-sale risk → Asset-Price & Balance-Sheet



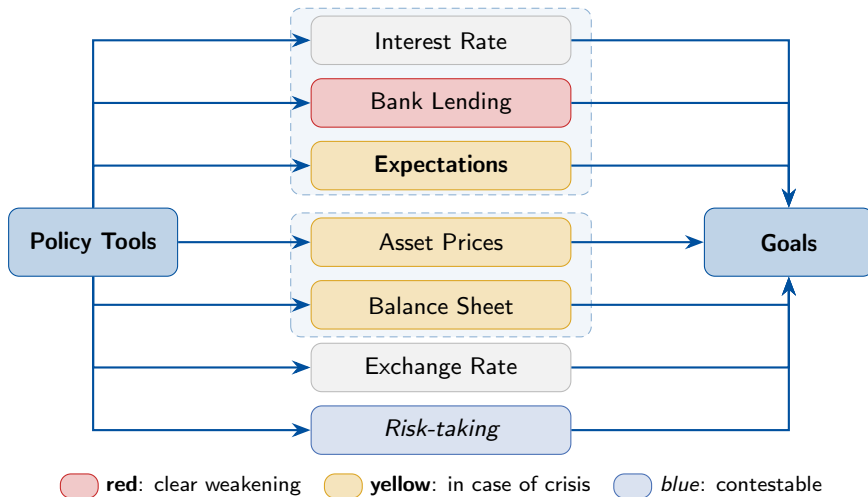
Chain: stablecoin crisis → T-bill fire-sale.

MiCAR vs GENIUS Act: A Closer Look

	MiCAR (EU)	GENIUS Act (US)
Philosophy	Stability, sovereignty	Innovation, USD reach
Scope	Comprehensive	Narrow
Issuers	EU banks / e-money institutions only	Banks + non-bank qualified institutions
Reserves	30% (60% for large) in bank deposits; rest in EU liquid assets	Flexible: T-bills, bonds; no deposit floor
Redemption	Guaranteed at par, no fees	Par redemption required; not government-backed, not federally insured, not legal tender

Reserve rules are not neutral — they constitute *industrial policy*.

Bailout pressure → the Expectations channel



DISINTERMEDIATION

Savers and borrowers bypass the regulated bank balance sheet, reducing the central bank's reach over credit conditions.

DekaBank context

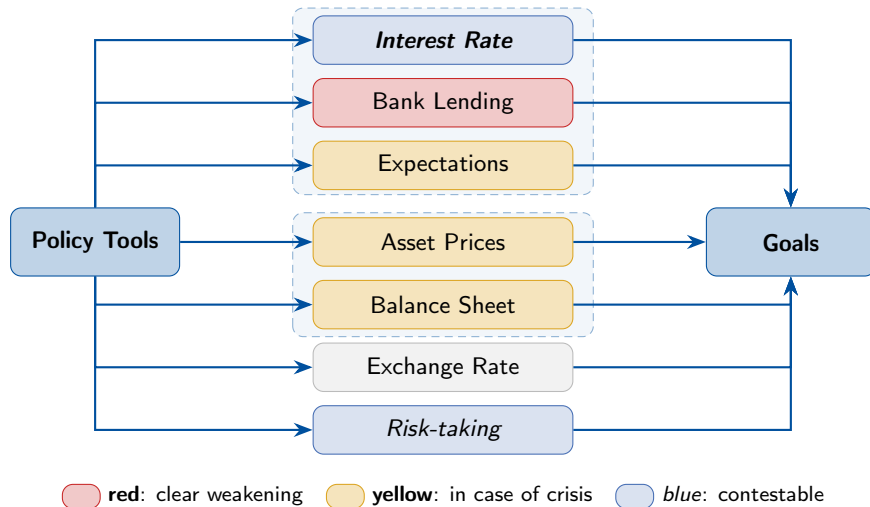
Investment arm of the German Sparkassen, €452 bn AuM (end-2025). Blockchain initiatives since 2018; first on-chain Pfandbrief (*BerlinHyp*).

Practitioner's view on disintermediation:

- Banks do **much more than intermediation**: settlement, custody, structuring, market-making, advisory.
- Real innovation: **atomic, programmable settlement** — infrastructure, not speculation.
- **Regulatory arbitrage is observable** — flows migrate to GENIUS-Act jurisdictions.
- **But tighter EU-framework is appreciated**

Interview, DekaBank (Frerker / Banh, April 2026).

Disintermediation and the Interest-Rate channel



CRYPTOMERCANTILISM

The strategic promotion of USD stablecoins to extend dollar dominance into on-chain financial infrastructure.

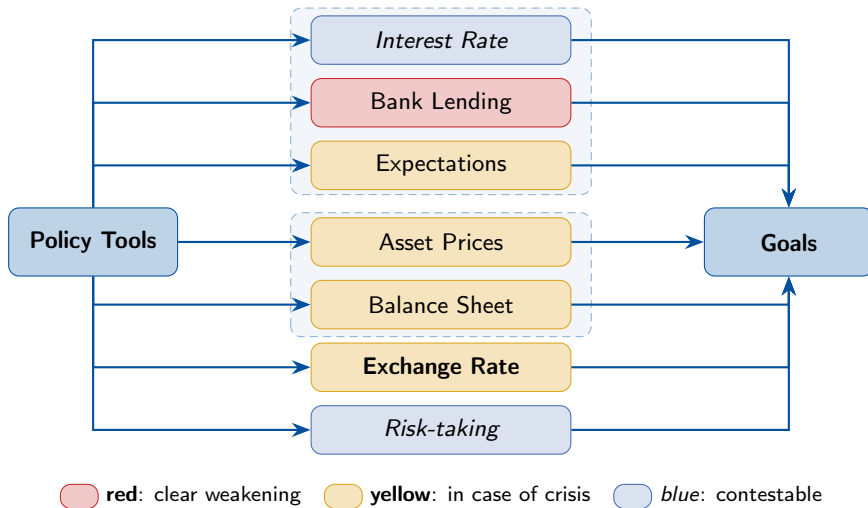
- US objective: extend the dollar's reserve status into **digital payment rails**.
- EU objective: prevent **digital dollarization** and preserve the euro's monetary role.

The EU's strategic aim

"De-dollarise crypto before crypto dollarises the EU."

European Parliament, Digital Assets & EU Framework (2025).

Digital dollarisation and the Exchange-Rate channel



Looking back — the honest answer

We started with one question

If money moves from bank deposits to crypto — what exactly breaks?

The heat-map answer

- **Red — one channel clearly weakened:**

- Bank lending — deposits leak, credit contracts regardless of the policy rate.

- **Yellow — four channels conditionally exposed:**

- Asset price — fire-sale risk on stablecoin reserves.
- Balance sheet — collateral chains hit by the same fire sale.
- Expectations — forward guidance under bailout pressure.
- Exchange rate — digital dollarisation as the rail grows.

- **Blue — two channels contested:**

- Risk-taking — arguably amplified, not weakened.
- Interest rate — redundant with bank lending; refuse to double-count.

Can central banks adapt fast enough — or is the 20th-century central bank an endangered institution?

Discussion thread 1:

Should the digital euro be accelerated?

Discussion thread 2:

Is CB independence sustainable in a populist era?

Thank you!

Questions?

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